Ian Douglas

Cloud Infrastructure & Security Engineer | AWS · GCP · Kubernetes

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PROFESSIONAL SUMMARY

Cloud infrastructure and security engineer with 12+ years designing, automating, and operating secure infrastructure across AWS, GCP, Azure, and on-premises data centers. At Stanford, operated multi-account AWS environments and a growing GKE footprint, moved the university's authentication stack (SAML/Shibboleth, Kerberos, OpenLDAP) onto Kubernetes, and ran multi-tenant Vault PKI. Builds reusable Terraform modules, CI/CD pipelines, and policy-as-code guardrails, and has supported research-scale workloads from petabyte-scale genomics to high-performance compute fleets for quantitative research.

TECHNICAL PROFICIENCIES

Cloud Platforms: AWS (Organizations, multi-account, EC2, VPC, Transit Gateway, Route 53, IAM, KMS, Secrets Manager, Lambda, EKS), GCP (GKE), Azure (AKS, VMSS), Hybrid & Multi-Cloud Integration

Infrastructure as Code & CI/CD: Terraform (reusable module libraries), GitHub Actions, GitLab Pipelines, Jenkins, GitOps (Argo CD, Flux CD), Git & Pull-Request Workflows

Networking: Hybrid Connectivity, Transit Gateway, Site-to-Site VPN, VPC Peering, Hybrid DNS, IPv6, IPsec, Firewalls, Certificates & PKI

Identity & Security: SAML/Shibboleth SSO, Kerberos v5, OpenLDAP, Active Directory, Microsoft Entra ID, AWS IAM, HashiCorp Vault (PKI), AWS KMS, SOPS, Policy-as-Code (OPA), NIST SP 800-207 Zero Trust, Least-Privilege Design

Containers & Platforms: Kubernetes (EKS, GKE, AKS, self-managed control plane & etcd), Docker, Custom Operators, Internal Developer Platforms

Observability: Splunk, Centralized Log Aggregation, Prometheus, Grafana, Fluentd, Elasticsearch

Languages & Automation: Python (Boto3, Pandas), Go, Bash, PowerShell, SQL, Agentic Development (Claude Code, Codex)

PROFESSIONAL EXPERIENCE

The Motley Fool | Alexandria, VA (Remote)

Senior Platform Engineer (Contract) | 04/2026 – Present

- **CI/CD Migration Automation:** Built reusable Python tooling that migrated ~200 repositories from Jenkins to GitHub Actions, including moving secrets from AWS Secrets Manager to dotenvx, saving months of manual effort.

- **Cloud Infrastructure:** Operate production AWS and Kubernetes infrastructure.
- **AI-Accelerated Engineering:** Developed the migration toolchain with Claude Code agentic workflows, shipping production-grade automation in a fraction of the usual time.

Kings Mountain Security | Palo Alto, CA (Remote)

Co-Founder, DevSecOps Engineer & Security Architect | 01/2025 – Present

- **Zero Trust Architecture:** Co-founded a security consultancy delivering Terraform-based "Zero Trust Capability Packages" modeled on NIST SP 800-207, turning security requirements into deployable guardrails and technical controls.
- **Compliance Automation:** Developing Security-as-Code modules that automate DoD Zero Trust compliance across the infrastructure lifecycle.

FlowDelta Trading | Boulder, CO (Remote)

Senior Software Engineer, Python/Data (Contract) | 01/2025 – 04/2026

- **Data-Intensive Analytics:** Built a backtesting platform and vectorized Monte Carlo engine that ran multi-dimensional parameter sweeps to validate trading algorithms.

ZagTech | San Jose, CA (Remote)

Senior Platform Engineer | 04/2024 – 12/2024

- **Golden Path Modules:** Architected reusable Terraform libraries for AKS and VMSS, establishing a standard path for compliant, self-service infrastructure provisioning.
- **Commercial Platform:** Engineered a proprietary Kubernetes platform deployed across multiple enterprise clients; the codebase was a primary value driver in the firm's subsequent acquisition.
- **Cloud Networking:** Designed Azure network connectivity for client environments, including VNet peering, site-to-site VPN, and hub transit routing.
- **Cost Optimization:** Rightsized and tuned Azure infrastructure to meet client cost requirements.
- **Documentation & Runbooks:** Wrote implementation guides and operational runbooks for client handoff and internal teams.
- **Developer Experience:** Built Go and Python tooling to automate GitOps workflows, increasing deployment velocity while reducing change failure rates.

Invitae Biosciences | San Francisco, CA (Remote)

Senior Platform Engineer | 07/2023 – 04/2024

- **Research Compute Platform:** Built a self-service platform on self-managed Kubernetes on AWS EC2 for petabyte-scale genomics workloads; executed zero-downtime cluster upgrades and stateful migrations.
- **Secrets Management Architecture:** Designed least-privilege secrets management using SOPS, AWS KMS, and Shamir secret sharing.

Pearson | London, UK (Remote)

Senior Cloud Platforms Engineer | 09/2019 – 08/2021

- **Multi-Account AWS:** Built and operated platform infrastructure across a multi-account AWS environment managed with AWS Organizations.
- **GitOps Platform:** Built and operated self-service developer platforms on Kubernetes with GitOps-based continuous delivery, including ephemeral Kubernetes environments for development and testing.
- **Custom Kubernetes Operators:** Wrote Go-based operators to orchestrate AWS services from self-managed Kubernetes on EC2.
- **Centralized Logging:** Implemented log aggregation with a distributed pipeline (Fluentd, Python, AWS Lambda) feeding Elasticsearch and Splunk for real-time analysis.

Stanford University | Palo Alto, CA

DevOps Engineer & Security Architect | 12/2015 – 09/2019

- **Identity Modernization:** Partnered with Infrastructure teams to move the university's authentication and authorization stack (SAML/Shibboleth SSO, Kerberos v5, OpenLDAP) from legacy servers onto Kubernetes.
- **Hybrid Active Directory Architecture:** Authored the reference architecture and Terraform code for Active Directory over IPv6 with transport-mode IPsec, certificate-based mutual authentication, and hybrid DNS, letting distributed teams in separate AWS VPCs join the domain securely without network peering or coordinated IP space; deployed through UAT into a small production footprint.
- **Hybrid Cloud Networking:** Built connectivity between on-premises infrastructure and AWS using Transit Gateway, site-to-site VPN, and VPC peering.
- **Multi-Account & Multi-Cloud:** Operated AWS Organizations multi-account environments and supported the university's growing GKE footprint alongside self-hosted Kubernetes.
- **Secrets Infrastructure:** Managed multi-tenant HashiCorp Vault clusters providing automated PKI, authentication proxies, and encryption as part of a three-person team.
- **Security Monitoring:** Integrated services with Splunk for centralized logging and security visibility.

Marvel Heroes (Gazillion Entertainment) | Foster City, CA

DevOps Engineer | 06/2014 – 12/2015

- **Cluster Ops Tooling:** Built PowerShell and SQL Server Management Objects (SMO) tooling to automate database operations and co-developed cluster operations tools for a flagship MMO with 40k+ daily active users, cutting service drain times by 93%.
- **Observability at Scale:** Operated large metrics and log aggregation servers for production game infrastructure.

Two Sigma | New York, NY

Systems Engineer | 05/2012 – 06/2014

- **High-Performance Fleet:** Automated bare-metal provisioning and configuration management for high-performance servers supporting research and trading operations.
- **Lifecycle Automation:** Orchestrated zero-touch deployment and lifecycle operations of 1,000+ secure workstations firm-wide.

Columbia University | New York, NY

Systems Engineer | 06/2010 – 05/2012

- **Infrastructure Operations:** Ran databases, ESX, storage, and VM infrastructure supporting Columbia's 22 libraries.

EDUCATION

University of Denver | Denver, CO

Bachelor of Arts (BA), Double Major: Mathematics & Philosophy

- **Mathematics:** Completed Bachelor of Science curriculum including grad-level coursework.
- **Philosophy:** Specialized in Formal Logic and Epistemology.